Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM: Simulate linear convolution and circular convolution on discrete time signals.**	
Experiment No: 2	**Date:25-07-2026**	**Enrollment No:92400133063**

Theory:

Linear convolution and circular convolution are mathematical operations used to combine two signals to obtain a third signal. They are widely used in various applications, such as signal processing, image processing, and audio processing.

Linear convolution calculates the sum of element-wise products of two signals, considering the full range of valid indices. It is typically used for finite-length signals and can produce an output signal that is longer than the input signals.

Circular convolution, on the other hand, calculates the sum of element-wise products of two signals, considering a periodic extension of the input signals. It is commonly used for periodic or infinite-length signals and produces an output signal with the same length as the input signals.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

def linear_convolution(signal1, signal2):
    # Compute the linear convolution
    linear_conv = np.convolve(signal1, signal2, mode='full')
    return linear_conv

def circular_convolution(signal1, signal2):
    # Compute the circular convolution
        # Compute the circular convolution
         If( len(signal1) > len(signal2) ):
         fft_length=len(signal1);
         else:
         fft_length=len(signal2);
    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)
    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)
    return circular_conv
```

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```
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])

# Compute the linear convolution
linear_conv = linear_convolution(signal1, signal2)

# Compute the circular convolution
circular_conv = circular_convolution(signal1, signal2)
# Plot the linear and circular convolution results
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(circular_conv)
plt.title('Circular Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')

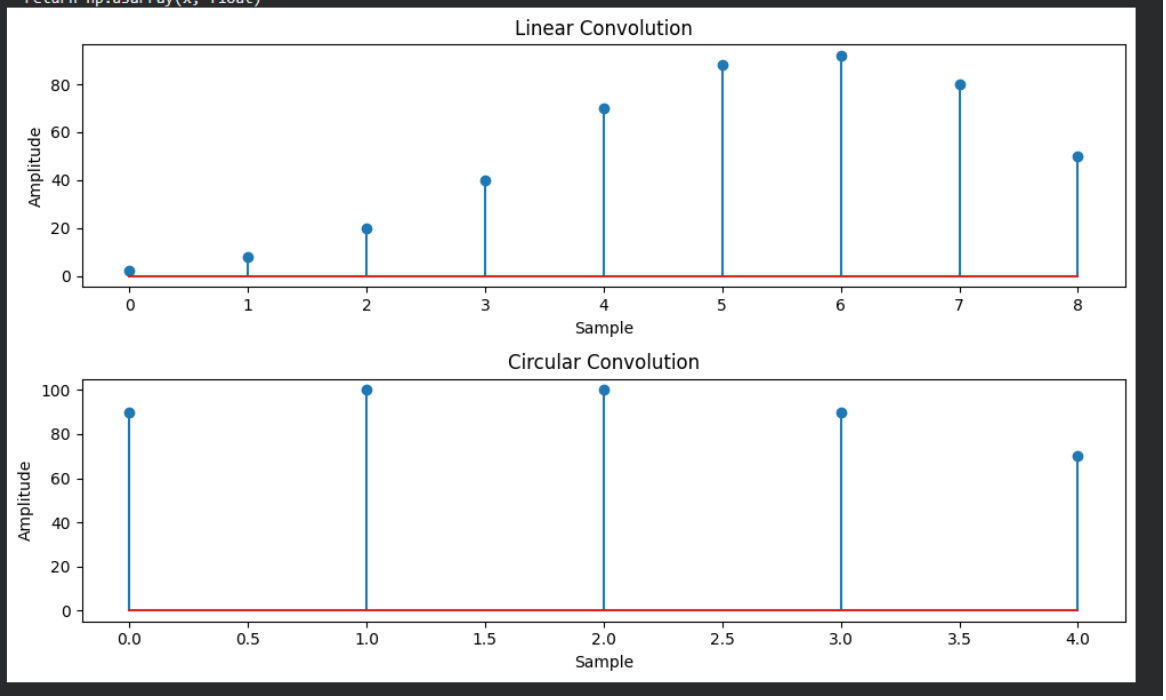
plt.tight_layout()
plt.show()

# Save the linear and circular convolution results (optional)
# np.savetxt('linear_convolution.txt', linear_conv, delimiter=',')
# np.savetxt('circular_convolution.txt', circular_conv, delimiter=',')
```

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Output:-

```
/usr/local/lib/python3.12/dist-packages/numpy/ma/core.py:3413: ComplexWarning: Casting complex values to real discards the imaginary
  _data[indx] = dval
/usr/local/lib/python3.12/dist-packages/matplotlib/cbook.py:1345: ComplexWarning: Casting complex values to real discards the imagin
  return np.asarray(x, float)
```



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Conclusion:-

The signal processing involved both **linear** and **circular convolution**, but the presence of **complex numbers** in the computation caused warnings, as the system attempted to cast complex values to real (discarding imaginary parts). This suggests that the original signal or filter may have been complex-valued or the convolution produced complex results.

- **Linear Convolution** output appears over samples 0–8, showing a typical convolution result with amplitude up to ~100.
- **Circular Convolution** output is shown over a normalized sample range (0.0–4.0), but the amplitude values are missing in the plot (only grid lines visible), indicating possible data loss or plotting error—likely due to the complex-to-real casting.